

DataCo

# Supply Chain Analytics Report

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*Identifying late delivery drivers · Optimising product profitability  
Enhancing fulfilment efficiency · Strategic recommendations*

Comprehensive analysis across supply chain, sales & customer data

# Problem Statement

“

*How can we leverage comprehensive supply chain, sales, and customer data to develop an analytics system that accurately identifies and predicts key factors contributing to late deliveries, optimises product and customer profitability, enhances fulfilment efficiency, and provides actionable insights and strategic recommendations to improve overall business performance and customer satisfaction?*



Late Delivery Prediction



Product & Profitability



Fulfilment Efficiency



Actionable Insights

# Objectives

Five analytical goals driving this supply chain study

- 1 Analyse Late Delivery Factors**  
Across shipping mode, region, market.
- 2 Identify Top Products**  
Identify which products drive the most sales, profit, and order volume.
- 3 Deep Dive Analysis**  
Uncover broader market trends and category-specific strategies.
- 4 Order Status distribution & Financial Impacts**  
Top 3 High-Risk Categories & financial impacts and mitigation strategies.
- 5 Build Predictive Model**  
Estimate whether an order is likely to face late delivery risk.

# Executive Summary

Five focus areas revealed by analysis of supply chain, sales, and customer data

54.8%

Average late delivery rate

## Late Deliveries

First Class promises 1-day delivery but consistently takes 2 days, driving a 95% late rate. Standard Class sits at 38%.

95%

First Class late rate

## Geographic Risk

USCA and Africa markets lead in late rates. Several countries (Armenia, Gambia, Eritrea) show 100% late delivery rates

39,832

Pending payment orders

## Payment Issues

All PENDING\_PAYMENT orders link exclusively to Payment\_Type\_ID 4 — a systemic gateway issue, not customer behaviour.

~2%

Sales lost to fraud

## Financial Risk

Top 3 categories lose ~2% of sales to fraud and cancellations. Fraud loss consistently exceeds cancellation rate across all three.



# Late Delivery Analysis

# Key Findings & High-Risk Country Root Causes

## Top Contributors to Late Deliveries

### 01 Specific Countries

Armenia, Gambia, and Eritrea show 100% late rates — logistics and infrastructure gaps.

### 02 Product Categories

Girls' Fashion, Camping & Hiking, Water Sports and Books show 100% late rates.

### 03 First Class Scheduling

Scheduled at 1 day but takes 2 days consistently — SLA is fundamentally unattainable.

### 04 Africa Market

Highest late delivery rate among all markets — limited shipping infrastructure and regional gaps.

## Why High-Risk Countries Struggle



### Logistics Challenges

Inadequate port and road infrastructure extends in-transit times beyond carrier estimates.



### Customs & Border

Slow customs clearance and stringent import documentation create unpredictable delays.



### Operational Capacity

Carrier capacity constraints and lack of local fulfilment centres increase lead times significantly.



### Economic / Political

Sanctions, currency instability, or regulatory barriers can halt or slow cross-border shipments.

# Shipping Mode Deep Dive: First Class vs Standard Class

## First Class

Late Rate: **95.27%**

Scheduled: **1 day**

Actual: **2 days**

Avg Delay: **+1 day**

## Standard Class

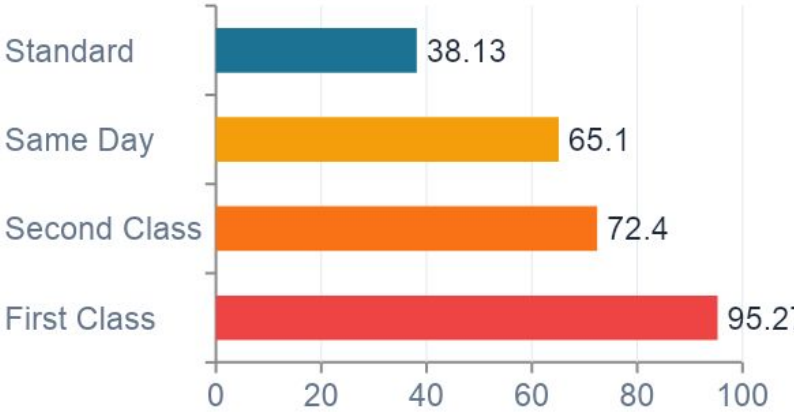
Late Rate: **38.13%**

Scheduled: **4 days**

Actual: **4 days**

Avg Delay: **~0 days**

Late Rate by Shipping Mode (%)



## Root Cause & Recommended Actions

**Unrealistic SLA:** First Class schedules 1-day delivery but logistics cannot support it — creating near-universal failure. **Revise SLAs:** Update First Class to 2-day promise to align with actual capability, improving customer satisfaction. **Standard Class:** 38% late rate still significant — investigate last-mile routing, carrier performance, and order complexity for this cohort.



# Product & Market Trending Analysis



# Top 10 Products by Total Sales

Products contributing the most to sales, profit, and order volume across the unified dataset

| #  | Product Name                                     | Total Sales    | Total Profit | Orders | Avg Unit Value |
|----|--------------------------------------------------|----------------|--------------|--------|----------------|
| 1  | Field & Stream Sportsman 16 Gun Fire Safe        | \$6,929,653.69 | \$756,220.77 | 15,164 | \$399.98       |
| 2  | Perfect Fitness Perfect Rip Deck                 | \$4,421,143.14 | \$493,828.30 | 20,359 | \$59.99        |
| 3  | Diamondback Women’s Serene Classic Comfort Bi... | \$4,118,425.57 | \$427,455.57 | 12,299 | \$299.98       |
| 4  | Nike Men’s Free 5.0+ Running Shoe                | \$3,667,633.20 | \$379,915.82 | 11,092 | \$99.99        |
| 5  | Nike Men’s Dri-FIT Victory Golf Polo             | \$3,147,800.00 | \$350,421.03 | 17,869 | \$50.00        |
| 6  | Pelican Sunstream 100 Kayak                      | \$3,099,845.09 | \$324,076.37 | 13,727 | \$199.99       |
| 7  | Nike Men’s CJ Elite 2 TD Football Cleat          | \$2,891,757.66 | \$311,902.82 | 18,783 | \$129.99       |
| 8  | O’Brien Men’s Neoprene Life Vest                 | \$2,888,993.91 | \$318,451.43 | 16,623 | \$49.98        |
| 9  | Under Armour Girls’ Toddler Spine Surge Runni... | \$1,269,082.67 | \$126,278.51 | 9,825  | \$39.99        |
| 10 | Dell Laptop                                      | \$663,000.00   | \$69,656.81  | 442    | \$1,500.00     |

# Key Insights — Top 10 Products

Strategic patterns derived from sales, profit, and order volume analysis

1

## Dominant High-Value Items

Field & Stream Gun Safe (\$6.93M) and Perfect Rip Deck (\$4.42M) lead — high-ticket items are critical revenue drivers.

2

## Consistent Profitability

High sales consistently correlate with high profit — indicating effective pricing and cost management across popular items.

3

## Sales vs. Order Volume

Gun Safe wins on unit value (15K orders), Rip Deck wins on volume (20K orders) — two distinct revenue models.

4

## Strategic Importance of Core Products

Nike footwear, Pelican kayaks, and O'Brien life vests are consistently strong performers — reliable revenue anchors that require continued inventory and marketing support.

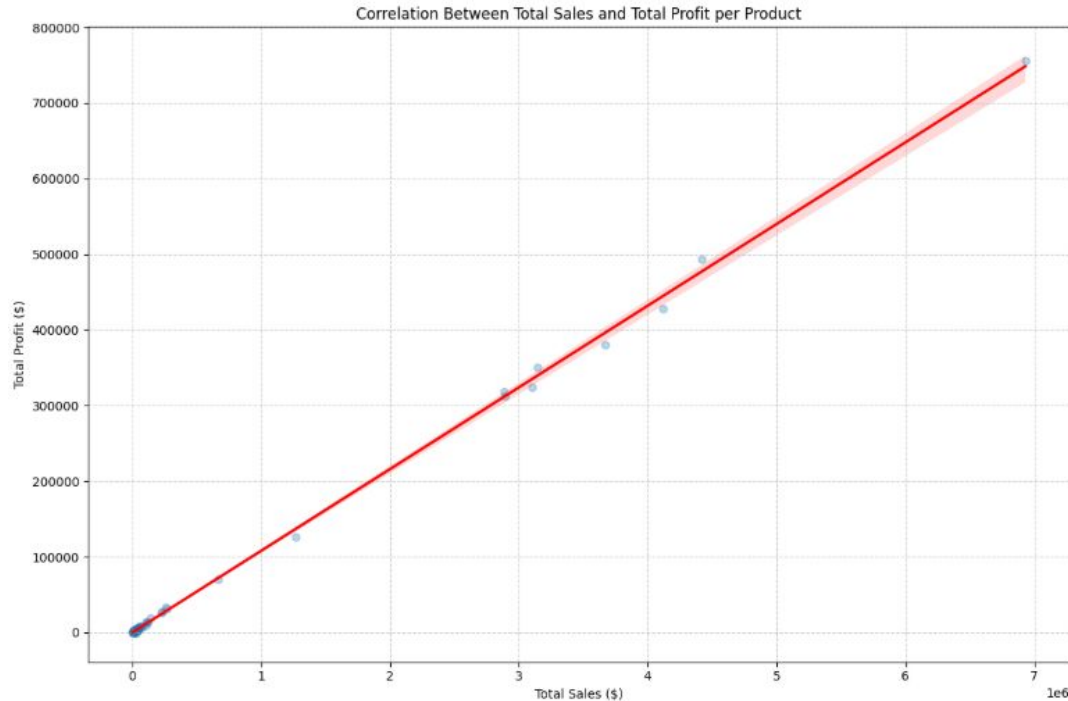
5

## Potential for Category Deep Dive

The diversity of top products points to an opportunity for deeper category-level analysis to uncover broader market trends and category-specific growth strategies.

# Correlation between Total Sales and Total Profit

The scatter plot above visually confirms a strong positive correlation between Total Sales and Total Profit. The upward-sloping regression line indicates that as Total Sales increase, Total Profit also tends to increase. This supports the observation that for most products, effective pricing and cost management are in place, leading to higher profitability as sales volume grows.



# Two Revenue Models - Findings and Strategic Recommendations

Derived from Top 10 Products analysis - Slide 9 data

## MODEL A | High Unit Value

Field and Stream Gun Safe | \$399.98/unit | 15,164 orders | \$6.93M

### Key Findings

- **Revenue through premium pricing, not scale.**

\$399.98/unit is 6.7x higher than Rip Deck. \$6.93M from just 15K orders - fewer fulfilment transactions needed.

- **Stronger per-transaction profit impact.**

\$756K profit at 10.9% margin. Premium buyers show low price sensitivity - protecting margins.

- **Seasonal, concentrated demand window.**

August peak from hunting season. Predictable demand spikes make inventory planning critical and achievable.

### Strategic Recommendations

-> Pre-stage inventory before Aug-Sep peak to prevent stockouts during concentrated demand windows.

-> Invest in premium fulfilment - expedited shipping and secure packaging - to justify the high price point.

-> Cross-sell accessories to grow basket size. High-value transactions maximise attach rate returns.

## MODEL B | High Order Volume

Perfect Fitness Rip Deck | \$59.99/unit | 20,359 orders | \$4.42M

### Key Findings

- **Revenue through market reach, not price premium.**

\$4.42M from 20,359 orders at \$59.99. 34% more orders than Gun Safe despite a 6.7x lower unit price.

- **Thin margins make fulfilment efficiency critical.**

\$493K profit at 11.2% margin. Shipping delays and returns erode thin per-unit margins quickly at volume.

- **Broad, repeat-purchase product with high elasticity.**

Fitness accessories at \$60 attract impulse and repeat buyers. Easily scalable across demographics.

### Strategic Recommendations

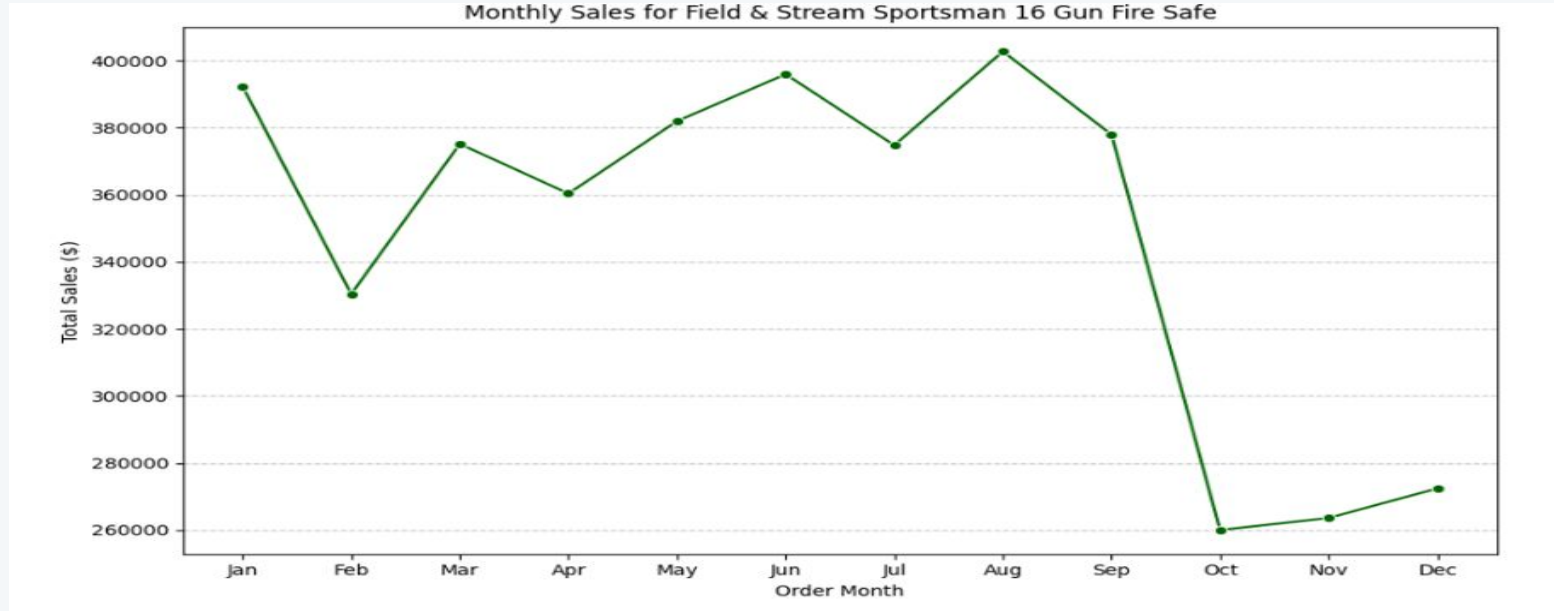
-> Optimise fulfilment efficiency. Standard Class and bulk dispatch reduce per-unit shipping cost to protect margins.

-> Bundle promotions and subscription offers to boost repeat purchase rate and customer lifetime value.

-> Scale via targeted digital marketing. Fitness influencer channels drive volume at low acquisition cost.

# Seasonal Analysis — Field & Stream Sportsman 16 Gun Fire Safe

Monthly total sales (\$) — seasonal demand patterns for inventory and marketing planning



## Peak: Aug ~\$420K

Hunting season & back-to-school overlap drive the highest monthly sales.



## Trough: Oct ~\$260K

Sharp post-summer drop signals need for targeted promotions in Q4.



## Inventory Implication

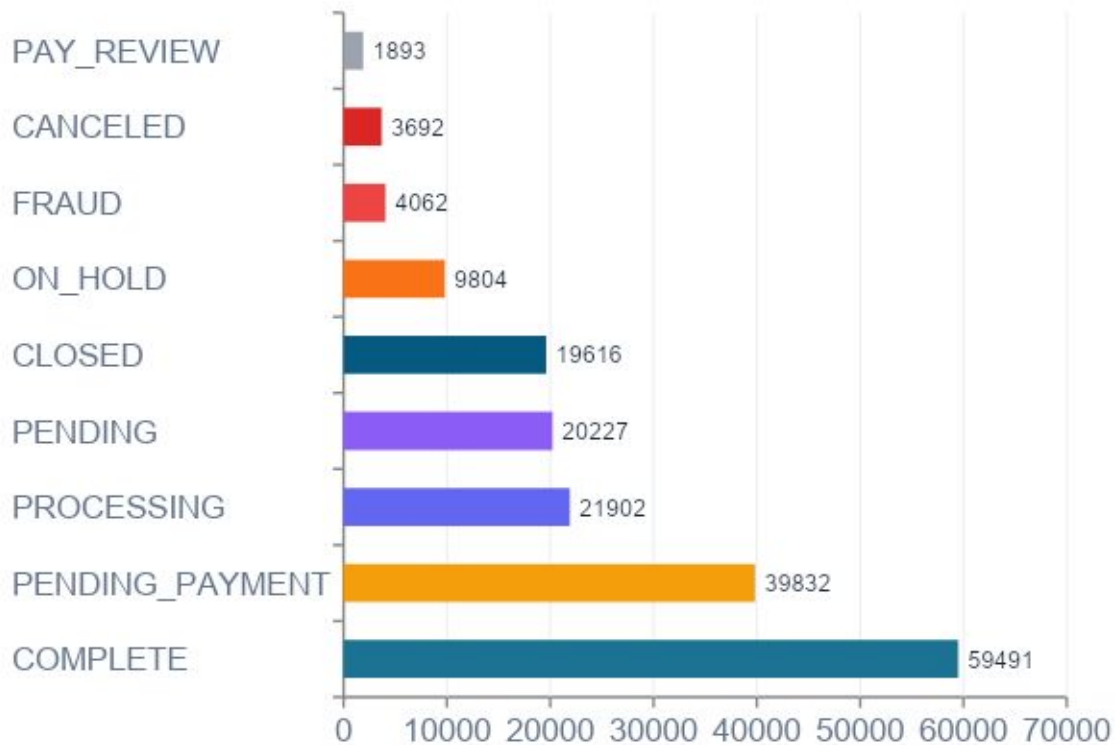
Pre-position stock in Jun–Aug; scale back in Oct–Dec to avoid overstock.



# Order Status distribution & Financial Impact

# Order Status Distribution

Analysis of 180,519 orders reveals fulfilment health and risk areas



**59,491**

Complete  
Orders ✓

**39,832**

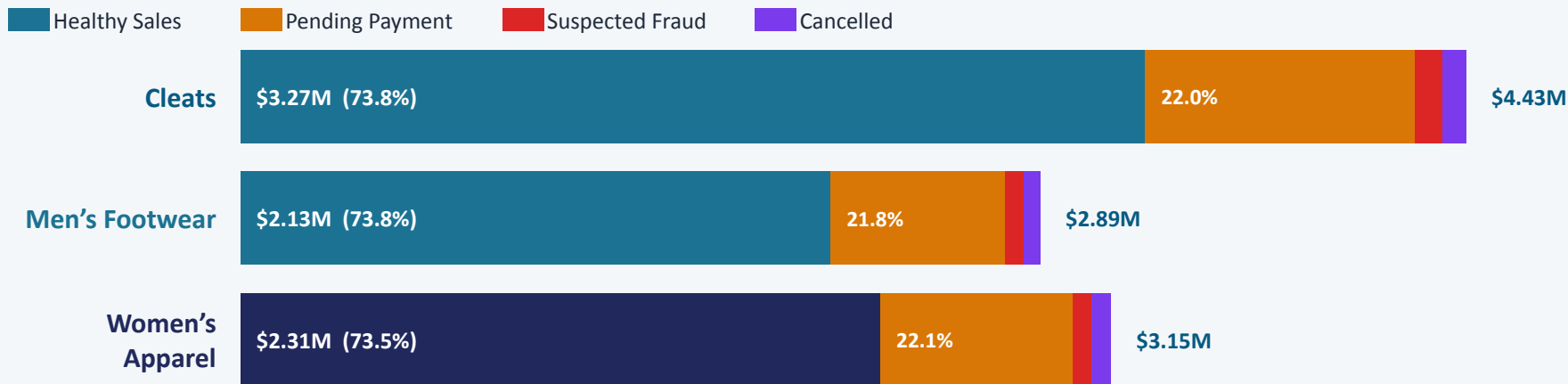
Pending  
Payment ⚠

**13,558**

High Risk  
(Fraud + Hold)

# Financial Impact — Top 3 High-Risk Categories

Sales exposure by order status — how each dollar of revenue is at risk



~74%

of revenue is healthy across all three categories

~22%

stuck in Pending Payment — a systemic gateway risk

~4%

permanently lost to fraud and cancellations combined



# High-Risk Category Loss Analysis

Sales lost to cancellations and fraud — dollar value, loss rate, and profit impact per category

| Cleats                                                                                                                                                                                                                          | Men's Footwear                                                                                                                                                                                                                | Women's Apparel                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Sales \$4,431,942.78                                                                                                                                                                                                      | Total Sales \$2,891,757.66                                                                                                                                                                                                    | Total Sales \$3,147,800.00                                                                                                                                                                                                    |
| <div><div>CANCELLED</div><div>1.94%</div><div>\$85,785.70</div><div>Profit lost</div><div>\$8,789.54</div></div> <div><div>FRAUD</div><div>2.29%</div><div>\$101,563.07</div><div>Profit lost</div><div>\$11,217.87</div></div> | <div><div>CANCELLED</div><div>2.10%</div><div>\$60,705.33</div><div>Profit lost</div><div>\$6,712.10</div></div> <div><div>FRAUD</div><div>2.32%</div><div>\$67,074.84</div><div>Profit lost</div><div>\$8,479.63</div></div> | <div><div>CANCELLED</div><div>2.09%</div><div>\$65,750.00</div><div>Profit lost</div><div>\$8,032.32</div></div> <div><div>FRAUD</div><div>2.24%</div><div>\$70,450.00</div><div>Profit lost</div><div>\$8,053.18</div></div> |
| Highest impact category                                                                                                                                                                                                         | Fraud dominates losses                                                                                                                                                                                                        | Balanced risk exposure                                                                                                                                                                                                        |

 **Key Insight:** Fraud loss rate consistently exceeds cancellation rate across all three categories. All figures hover near ~2% per risk type — a systemic pattern, not isolated incidents.

# Recommended Analytical Next Steps

## Investigate PENDING\_PAYMENT



Examine payment methods, demographics, order values, and temporal patterns to identify root causes of payment failures.

## Analyse High-Risk Statuses



Quantify financial impact of ON\_HOLD, FRAUD, CANCELED, and PAYMENT\_REVIEW orders. Map to specific products, segments, or geographies.

## Evaluate Fulfilment Speed



Measure average time in PROCESSING and PENDING states. Identify optimisation opportunities to reduce cycle times.

## Correlate with Late Deliveries



Determine if PENDING status orders are more likely to become late deliveries — enabling proactive intervention.

## Customer Behaviour Analysis



Analyse fraud and cancellation patterns to improve risk scoring and detection models for high-risk customers.

# PENDING\_PAYMENT Deep Dive



## Critical Finding:

All PENDING\_PAYMENT orders are exclusively linked to Payment\_Type\_ID 4 — suggesting a systemic payment gateway failure rather than individual customer issues.

## Top Affected Countries



**Estados Unidos**

Highest volume



**Francia**

High concentration



**México**

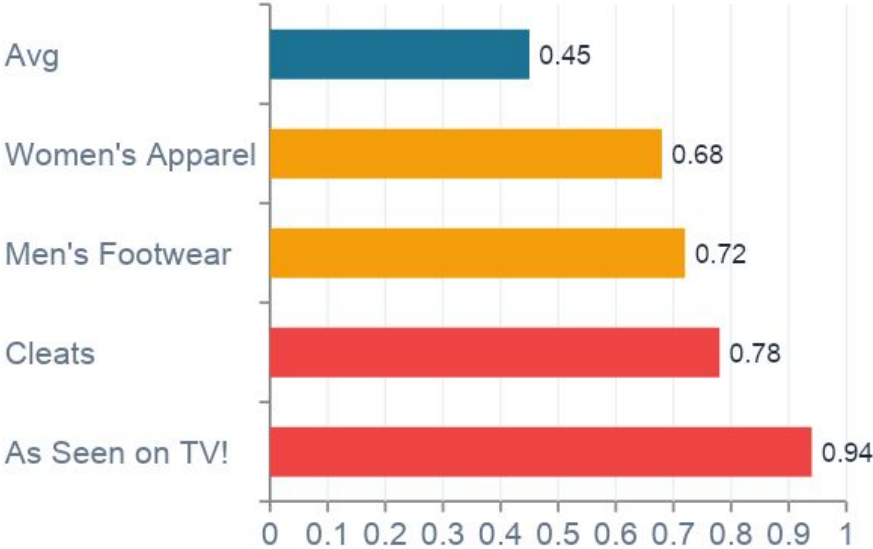
Regional priority



**Other Markets**

Systemic gateway issue

Pending-to-Complete Ratio by Category



# PENDING\_PAYMENT Mitigation Strategies

*Target: High-risk categories — Cleats, Men's Footwear, Women's Apparel | All linked to Payment\_Type\_ID 4*

1

## Enhanced Gateway Monitoring

Implement real-time monitoring for Payment\_Type\_ID 4 transactions on high-risk categories. Identify and fix integration errors causing silent failures.

### Goal

Reduce PENDING volume at the gateway level — addressing root cause rather than symptoms.

2

## Proactive Customer Outreach & Alternatives

For orders stuck in PENDING\_PAYMENT >24h, trigger automated communications with alternative payment options (card, wallet, etc.).

### Goal

Convert stalled payments into successful transactions — recovering lost sales and profit.

Action Priority: High — every PENDING\_PAYMENT order represents unrecovered revenue and customer friction

# Product Optimisation to mitigate Financial Impact

Four targeted actions for Cleats, Men's Footwear, and Women's Apparel to reduce cancellations, fraud, and fulfilment failures

1

## Product Info & Standardisation

Audit and standardise descriptions, images, and sizing guides. Add 360° views, measurements, and virtual try-on tools for Cleats, Footwear, and Apparel.

**Impact:** Reduce cancellations from expectation gaps — cutting reverse logistics costs.

2

## Post-Cancellation Feedback Loop

Mandatory quick survey for cancelling customers. Analyse themes — pricing, availability, shipping — and link findings to inventory and product management systems.

**Impact:** Address root causes and improve product availability in key regions.

3

## Inventory Placement Strategy

Pre-position stock for high late-rate categories closer to top markets (US, France, Mexico). Implement real-time inventory tracking for high-risk products.

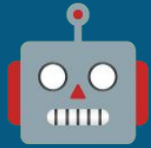
**Impact:** Minimise stockouts and reduce transit times, especially for premium shipping.

4

## Discount Strategy Review

High discount rates correlate strongly with late delivery risk. Re-evaluate promotions for operational feasibility. Cap volumes on deeply discounted items during peak periods.

**Impact:** Decouple revenue strategy from operations to reduce fulfilment failures.



# Prediction Model & Feature Importance

# Predictive Model Performance

Classification model trained on 32,825 orders — strong discriminative power for late delivery risk

73%

Overall Accuracy

0.81

ROC AUC Score

79%

Late Risk Precision

69%

Late Risk Recall

## Confusion Matrix

- ✓ True Negatives (on-time correctly predicted): 11,600
- ✓ True Positives (late correctly flagged): 12,441
- ⚠ False Negatives (late missed): 5,540
- ⚠ False Positives (false alarms): 3,244

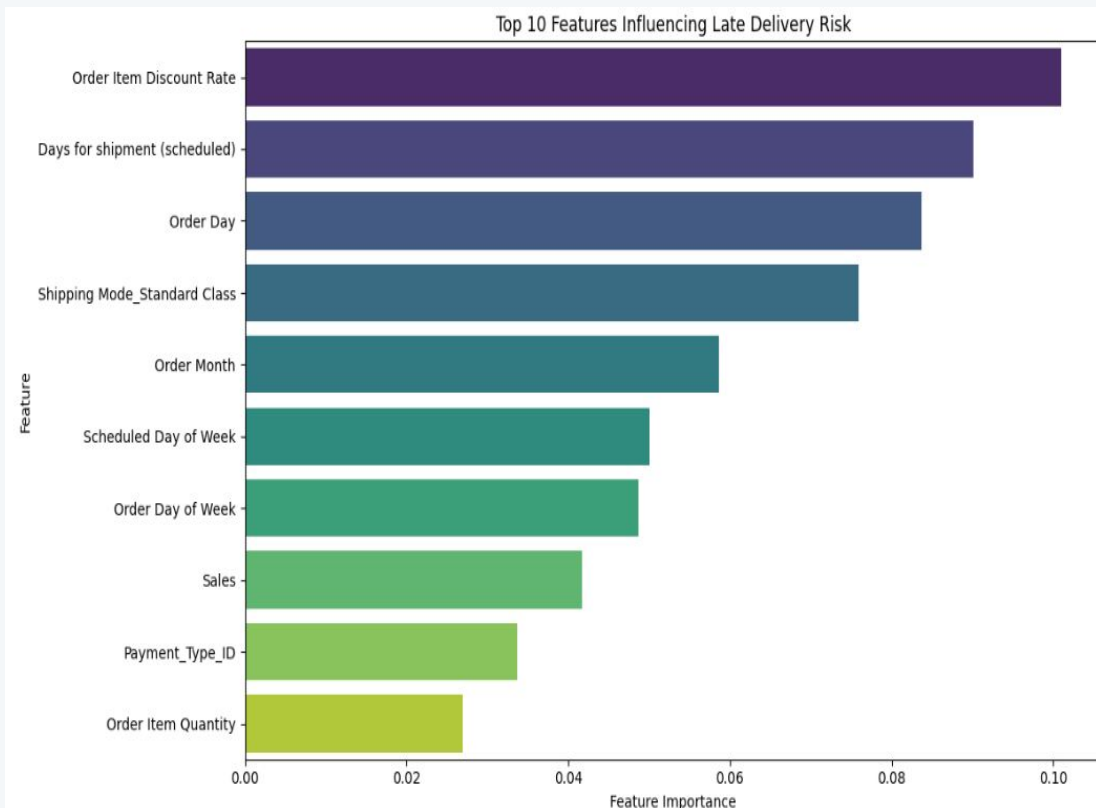
## Operational Value

Out of every 100 total orders processed by the model, **73 are correctly** flagged as a late delivery risk or correctly identified as arriving on time

When the model flags an order as late risk, it is correct **79% of the time** — a highly reliable signal for operations to act on proactively.

AUC of **0.81** confirms strong discrimination power beyond chance.

# Late Delivery Prediction — Key Features



## #1 Discount Rate

High discounts drive late risk — possible stockout or slow pick/pack correlation.

## #2 Scheduled Days

Unrealistic SLAs (especially First Class) are a primary predictor.

## #3 Standard Class

38% late rate — routing, last-mile, and order complexity factors.

## Time Features

Order day, month, and scheduled weekday are significant — seasonal peaks exist.



# Feature Importance — Operational Actions

Top model predictors translated into business recommendations

#1 —  
10.1%

## Promotional and Discount Strategy (Discount Rate)

High discounts trigger volume spikes that overwhelm fulfilment capacity. **Recommendations:** Cap daily order volumes for deeply discounted items; add 1–2 buffer days to scheduled delivery when discount codes are applied; pre-stage inventory before sale events.

#2 — 9.0%

## Shipping Mode Optimisation (Scheduled Days + Standard Class)

Scheduled windows are too rigid and don't account for real carrier variability. **Recommendations:** Audit Standard Class carrier SLAs and shift volume to regional carriers on high-delay lanes; revise First Class SLA to 2 days to match operational reality.

# Strategic Recommendations

01

Revise First Class SLA to 2 days to align with operational reality and reset customer expectations.

02

Deep-dive audit of Payment\_Type\_ID 4 gateway. Partner with provider to fix silent failure mode.

03

Suspend or review shipping to 100%-late-rate countries. Engage regional logistics partners.

04

Pre-position inventory near top markets (US, France, Mexico) for high-risk and discounted SKUs.

05

Implement predictive late-delivery model to flag at-risk orders at point of placement.

06

Enhance product content for Cleats, Footwear, Apparel to cut cancellations and reverse logistics.

# Thank You

**Freddy Fok**

*Questions & Discussion*

DataCo Supply Chain Analytics · Comprehensive analysis of late deliveries, payment issues, product optimisation & predictive modelling